

ASSIGNMENT 1: ANCIENT CRYPTO

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1 The Index of Coincidence (IC)

[20 pts] Question 1 For each of Caesar, affine, Vigenère, and Playfair, is the IC of the ciphertext expected to be higher, lower, or equal to the plaintext's IC? Explain why.

Caesar: equal

Explanation: Caesar cipher shifts every letter by a fixed number. Since each plaintext letter is mapped to exactly one ciphertext letter, the letter frequencies remain unchanged, and therefore the IC stays the same.

Affine: equal

Explanation: Affine cipher uses a one-to-one substitution, mapping each plaintext letter to a unique ciphertext letter. This preserves the letter frequency distribution, so the IC remains unchanged. This mapping is a permutation of the alphabet.

Vigenère: lower

Explanation: Vigenère cipher uses different shifts depending on the position of the letter in the repeating key. Therefore, the same plaintext letter can be encrypted into different ciphertext letters, making the letter frequency distribution more uniform and generally reducing the IC. If the key length is 1, Vigenère reduces to Caesar and the IC is unchanged.

Playfair: lower

Explanation: Playfair cipher encrypts pairs of letters rather than individual letters. The encryption of each letter depends on the other letter in its pair. This disrupts the original single-letter frequency distribution, making it generally more uniform and reducing the expected IC.

2 Detecting Ciphers

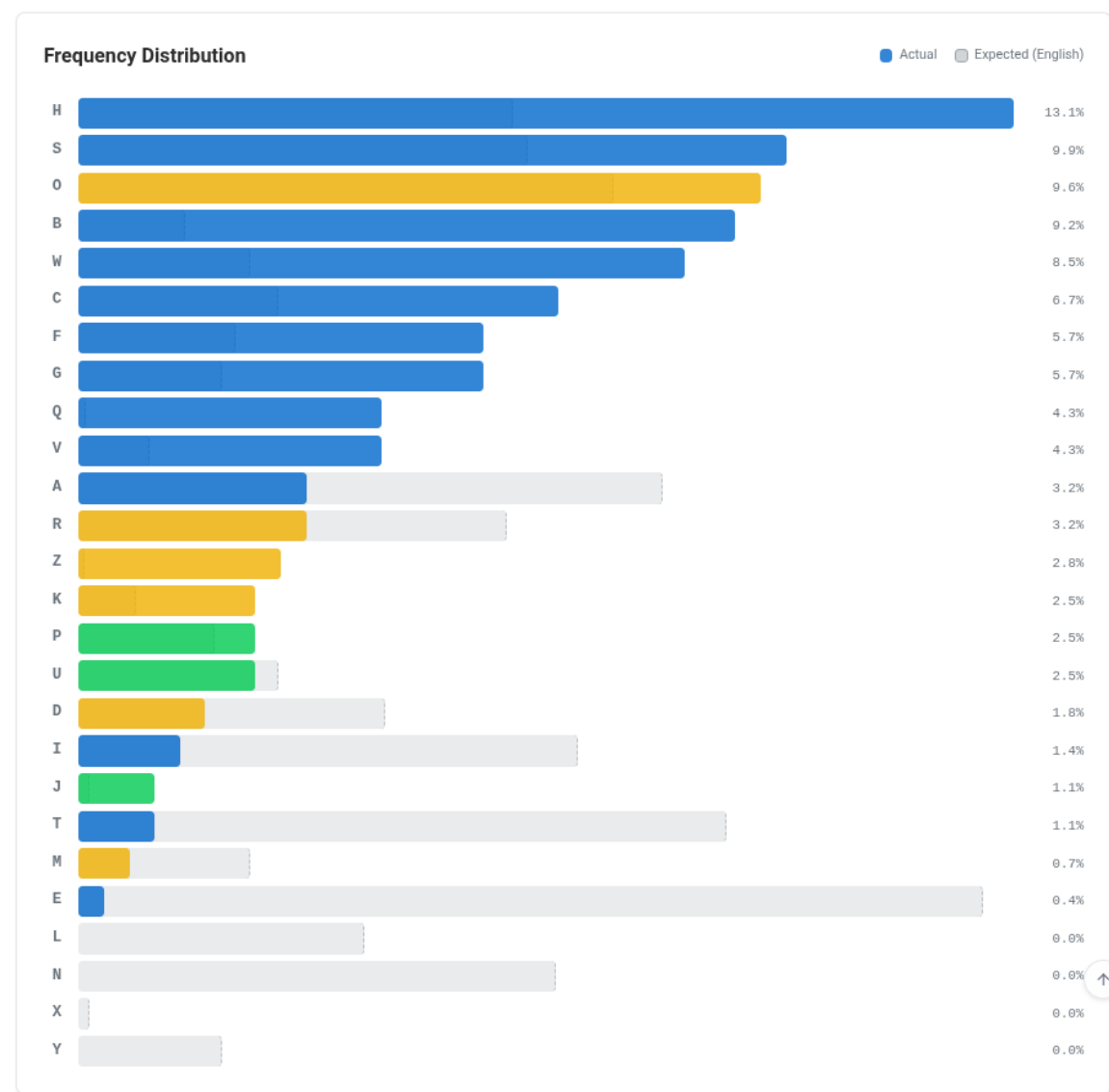
[30 pts] **Question 2** Without breaking them, which line of `ciphertexts_i.txt` corresponds to which cipher? Explain how you figured out each one.

The six lines of `ciphertexts_80.txt`:

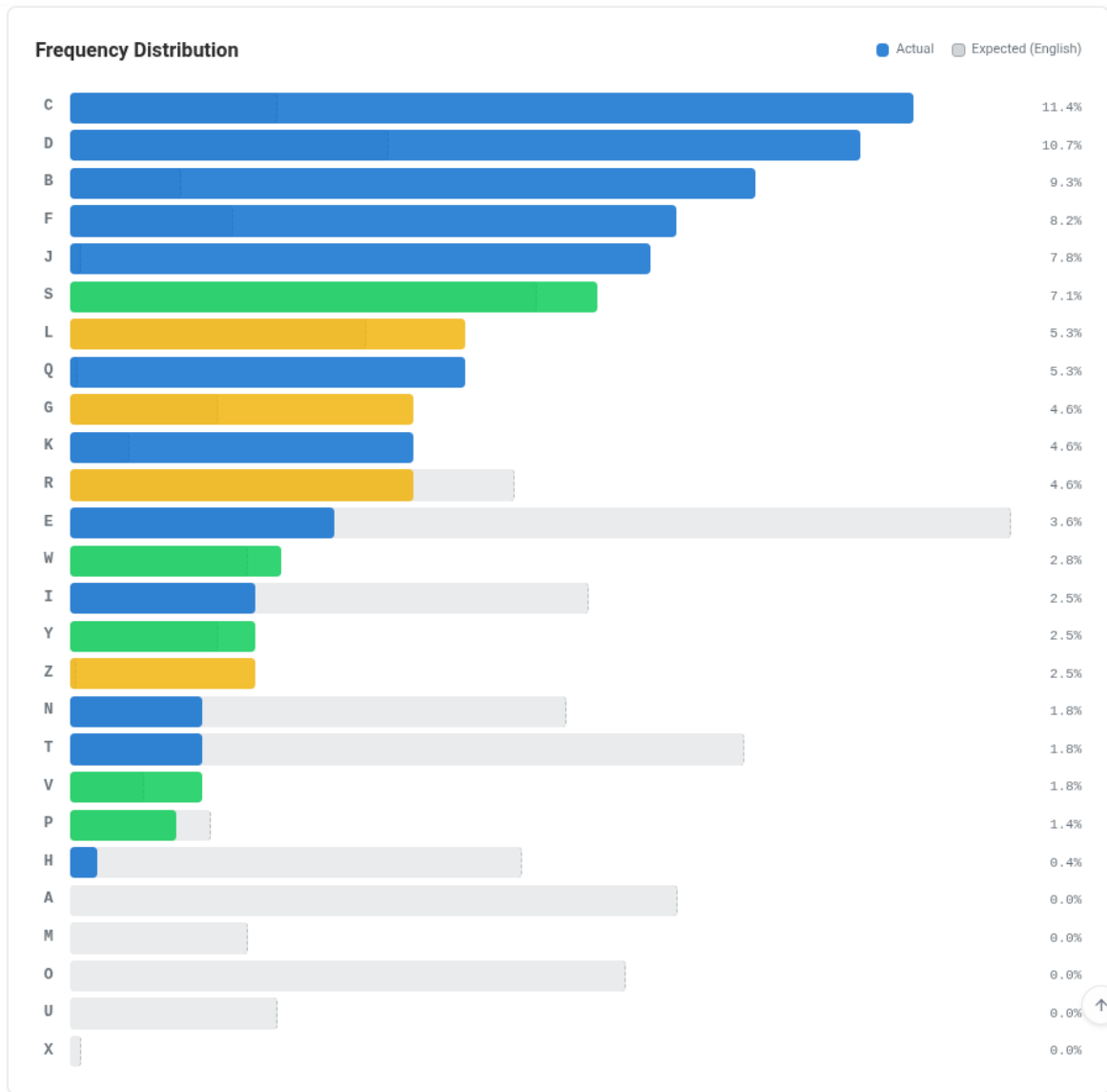
- 1) PRWAKPWAGBYBHDWPSNQPOWKZZQYCINQDQMAHFZMQMUZPDPIUMQLFNBRLONCYCHUQBRLPGBOGPRGDPTVQWHABAGPWQBQHVHVMG
KFRGXPKMGPHUVYLMFLSMBNKFHVVMQTDPSYCWBUMKXKFZNQZRXHWCLBWHRQDVUMFTQKLDPVCFDOBUIBQGCXHBQSRESMYOMUKDPYPDPT
VDPNUXQQBBEHPNCFZQZKQZNQLONCHWKOLUZGPPYVBUMQHKONVSURXPCSFUVUMQTDPSDPZAOGUMMQHTBOHGHDXKNVGPHTMQMU
- 2) GZCKPFCAPCRWSGHFBOBGWHWCBOBRAOHIFWHMKWHVWHGQOZARSASOBCFOBRJJCZJWBUOPWZWHWSGWHRSAACBGHFOHSGHVHOFCKHVF
SEIWFSGPCHVORODHOHWCBOBROQQSDHOBQSGZCKPFCHSOQVSGHVHQBQVUBUSQOBPFWBUBSKGHFSBUHVGVWHFSTZSQHGHSWADCFHOBQSC
DOHWSBQSOBRHWAWBUGVCKWBUHVOHHFISRSJSZCDASBHPQOZOBQSGQCBHWBIWHMKWHVHFOBGTCFAOHWC
- 3) B3P3R6IFG5Q7SOUUMDKFFFMPZ2SKKXHYT2F6DZB2EFSXDTI3WEI542Q4X4UYYPVMID0ZW4MBXXDUGNGNEILQLXNMADQQ55YIQT3J
D5Y4QOLOPJRMISS3OCV2YENQFMJP4IAJINJJP7LYM2RWQIKZEZL4K7PFVYJOS7EVWQ6AL4T6B34HHJ2OC7PFTNCBHSQL4VYPBIMGYKZ
6RK3BEMXP5T2ZAWBTTJKOAKBWFNUCCGHG7YOFBTVJT5H4FA76JU5TPZM4XP24GAFLQUANIWGV7CCLTS02LIQAL2PIBGSOWIWKYKQCHY
ZLE2
- 4) F6ADXSKGN5QCHHVCOUJ4PIIVKPDJYKIOUQS34SWWXS3ISIO3MIEXSQZ2JYDNAV6XVOZWAM6HNQFCTQMXVR65RRRLRVT3UHN4P2J
D7VTABKYDLHA7YK5DIEND3DHG06L20JCEEBTPS6YKVD4RWCTMVH7DNYDXZOMCOEQHFLBMQ4OBNAARJPIPPNEFEJUFVDA3D6X6GKSMH
SC7LEPUVAJLLXRC04SNR5KKB5OE7UTVCZAUJFTHNZK2LFW2KJXWAFPKXU4WJRX56WKUNI
- 5) NFYVCLEJQDEQDJDSCJEQDRBJBLSFSKJNBGGHBCIJIFQEGBZYJFSKWLVRJDKZLCBLSBCKDZLSJCQFCDJZFJCDQTCIQLVPIEQFRCBR
DFSKQDWBSDZDSCNFYVCLEJCDFRIDJCIFCRLSJBBCDSCDWLQCFSKFKFECFYGBCTGDFKCLDWBRBDSRTBCQDWGDRCJKBJRBEGBSDFPB
GBCTFSKJCQFCDPBRREFYFGBCTNFYVCLEJDZYLKBDJJNBGGQDWBSDZDSCFSKRLSCQLGGDKJCQDSPCI
- 6) CUGFEKEHEEBFTCEKCFBFATOIQNIQNRDXSQNVHAYXOICYDXRFCGUDTZFEMNZJMILDVCINKTIQNEKTYUJOOEIEZCGOITCNSTZEHMS
GEUNOFTVQWHWSKXUTLERCQJRCJLAFZUMCSPPRCLATPYNRWFUCWTKTYUCMHOIJUNUEFVWODLVSNINEVVZJOTRDXADIXDGFTJXIWAN
XJBALCFVIFEUVICMSJJLEFGKXYNKIEVFUWNTUUNVEWVYCLIMUHEKSDQANWTFDYMTOUYYSMNZJSEXFZSCEFCPQHDKTIQNEYITSIOJDZ
DUTAOE

For this question, the ciphers are identified in this order: OTP first, then Playfair, then Vigenère, then Caesar, and finally Affine.

These two figures are the letter-frequency plots of the remaining lines, line 2 and line 5.



Line 2. The ciphertext was pasted into <https://caesarcipher.org/ciphers/frequency-analysis>, which drew this chart.



Line 5. Same tool, <https://caesarcipher.org/ciphers/frequency-analysis>.

Caesar: line 2

Explanation: Caesar shifts every letter by the same x . Guess which bar is E; then T, A, O, ... are that letter plus the same x . A red cell is under 1%, so that guess is wrong.

Line 2. $E \rightarrow H$ ($x = 3$) sends I to L (0%), so it is wrong. $E \rightarrow S$ ($x = 14$) keeps E through D high, so line 2 is Caesar with shift 14.

Image of E	x	E	T	A	O	I	N	S	H	R	D
H (13.1%)	3	H 13.1%	W 8.5%	D 1.8%	R 3.2%	L 0.0%	Q 4.3%	V 4.3%	K 2.5%	U 2.5%	G 5.7%
S (9.9%)	14	S 9.9%	H 13.1%	O 9.6%	C 6.7%	W 8.5%	B 9.2%	G 5.7%	V 4.3%	F 5.7%	R 3.2%

Line 5. Every row has a red cell, so no single shift fits. Line 5 is not Caesar.

Image of E	x	E	T	A	O	I	N	S	H	R	D
C (11.4%)	24	C 11.4%	R 4.6%	Y 2.5%	M 0.0%	G 4.6%	L 5.3%	Q 5.3%	F 8.2%	P 1.4%	B 9.3%
D (10.7%)	25	D 10.7%	S 7.1%	Z 2.5%	N 1.8%	H 0.4%	M 0.0%	R 4.6%	G 4.6%	Q 5.3%	C 11.4%
B (9.3%)	23	B 9.3%	Q 5.3%	X 0.0%	L 5.3%	F 8.2%	K 4.6%	P 1.4%	E 3.6%	O 0.0%	A 0.0%
F (8.2%)	1	F 8.2%	U 0.0%	B 9.3%	P 1.4%	J 7.8%	O 0.0%	T 1.8%	I 2.5%	S 7.1%	E 3.6%
J (7.8%)	5	J 7.8%	Y 2.5%	F 8.2%	T 1.8%	N 1.8%	S 7.1%	X 0.0%	M 0.0%	W 2.8%	I 2.5%
S (7.1%)	14	S 7.1%	H 0.4%	O 0.0%	C 11.4%	W 2.8%	B 9.3%	G 4.6%	V 1.8%	F 8.2%	R 4.6%
L (5.3%)	7	L 5.3%	A 0.0%	H 0.4%	V 1.8%	P 1.4%	U 0.0%	Z 2.5%	O 0.0%	Y 2.5%	K 4.6%
Q (5.3%)	12	Q 5.3%	F 8.2%	M 0.0%	A 0.0%	U 0.0%	Z 2.5%	E 3.6%	T 1.8%	D 10.7%	P 1.4%
G (4.6%)	2	G 4.6%	V 1.8%	C 11.4%	Q 5.3%	K 4.6%	P 1.4%	U 0.0%	J 7.8%	T 1.8%	F 8.2%
K (4.6%)	6	K 4.6%	Z 2.5%	G 4.6%	U 0.0%	O 0.0%	T 1.8%	Y 2.5%	N 1.8%	X 0.0%	J 7.8%
R (4.6%)	13	R 4.6%	G 4.6%	N 1.8%	B 9.3%	V 1.8%	A 0.0%	F 8.2%	U 0.0%	E 3.6%	Q 5.3%

Affine: line 5

Explanation: Line 5 is the other monoalphabetic line, and the table shows it is not a fixed shift, so it is Affine.

Vigenère: line 6

Explanation: After identifying Playfair (line 1) and OTP (lines 3 and 4), the remaining candidates are lines 2, 5, and 6. Caesar and Affine are monoalphabetic, so their IC should stay close to English. Vigenère is polyalphabetic, so its IC should drop toward a random string.

Steps.

1. Compute the unnormalized IC

$$\kappa = \sum_i \frac{n_i(n_i - 1)}{N(N - 1)}.$$

2. Multiply by 26 so the values can be compared with the figures on Wikipedia (https://en.wikipedia.org/wiki/Index_of_coincidence): English ≈ 1.73 , uniform random = 1.00.

3. Results for the three remaining lines:

- Line 2: $0.06878 \times 26 = 1.79$
- Line 5: $0.06472 \times 26 = 1.68$
- Line 6: $0.04308 \times 26 = 1.12$

4. Lines 2 and 5 are both near 1.73, so they are Caesar or Affine. Line 6 differs most from 1.73 and is closest to 1.00, therefore it is Vigenère.

Playfair: line 1

Explanation: Line 1 is the only ciphertext that contains no letter J. In this assignment, Playfair replaces every J with I and uses a 25-letter key that excludes J, so a

Playfair ciphertext can never contain J. The other letter-only lines (2, 5, and 6) all contain J, so they cannot be Playfair.

OTP: lines 3, 4

Explanation: Lines 3 and 4 are the only ciphertexts that contain digits. The OTP in this assignment encodes letters as 5-bit symbols with the RFC 4648 Base32 alphabet (A–Z and 2–7), XORs them with the pad, and encodes the result back with the same alphabet. Therefore an OTP ciphertext can include the digits 2–7, whereas Caesar, Affine, Vigenère, and Playfair output only A–Z. This also matches the note that OTP was used twice.

3 Breaking Ciphers

[50 pts] Question 3 For each ciphertext, give the name of the Pokémon described in the plaintext and explain how you found it.

Caesar: Slowbro

Explanation: Question 2 gave the key $x = 14$. Subtract 14 from every letter. The plaintext begins with SLOWBRO.

SLOWBROEMBODIESTRANSITIONANDMATURITYWITHITSCALMDEMEANORANDEVOLVINGABILITIESITDEMONSTRATESTHATGROWTHREQUIRESBOTHADAPTATIONANDACCEPTANCESLOWBROTEACHESTHATCHANGECANBRINGNEWSTRENGTHSITREFLECTSTHEIMPORTANCEOFFATIENCEANDTIMINGSHOWINGTHATTRUEDEVELOPMENTBALANCESCONTINUITYWITHTRANSFORMATION

Affine: Kabutops

Explanation: Encryption is $ax + b \equiv y \pmod{26}$, so $ax \equiv y - b \pmod{26}$. Decryption needs a^{-1} , and that exists only when $\gcd(a, 26) = 1$. The 12 possible values of a are

$$1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25.$$

The tallest ciphertext bars are C, D, B, F, and the common English letters are E, T, A, O. Pair them and solve. $E \rightarrow D$ and $T \rightarrow C$ give

$$\begin{aligned} 4a + b &\equiv 3 \pmod{26}, \\ 19a + b &\equiv 2 \pmod{26}. \end{aligned}$$

Subtract: $15a \equiv 25 \pmod{26}$. Since $15 \cdot 7 \equiv 1$, $a \equiv 25 \cdot 7 \equiv 19$. Then $4 \cdot 19 + b \equiv 3$, so $b \equiv 5$.

With $a = 19$ and $b = 5$, the encryption formula is

$$19x \equiv y - 5 \pmod{26}.$$

Multiply both sides by a number k :

$$19kx \equiv k(y - 5) \pmod{26}.$$

The left side should be only x , so k must satisfy $19k \equiv 1 \pmod{26}$. Then

$$19kx \equiv 1 \cdot x \equiv x \pmod{26}.$$

$19 \cdot 11 = 209 = 8 \cdot 26 + 1$, so $k = 11$. Each ciphertext letter y becomes $11(y - 5) \pmod{26}$. The plaintext begins with KABUTOPS.

KABUTOPSREPRESENTSPRECISIONANDSKILLWITHSHARPLIMBSANDFOCUSEDMOTIONITDEMONSTRATESMASTERYTHROUGHPRACTICEAND
DREFINEMENTKABUTOPSTEACHESTHATCONSISTENTEFFORTANDADAPTABILITYLEADTOEFFICIENCYITREFLECTSDISCIPLINEAGILIT
YANDSTRATEGICCAPABILITYKABUTOPSEMBODIESKILLREFINEMENTANDCONTROLLEDSTRENGTH

Vigenère: key length 6

Explanation: Line 6 has 312 letters. Its IC is 1.120, near a random string. For a guessed key length k , group i is letters $i, i + k, i + 2k, \dots$. Those letters share one key letter, so a correct k makes each group's IC near English, about 1.73. The groups below start at $k = 3$.

Key length 3. 104 letters each.

1. CFEETKBTQGDQHXCDXDFMJLCKQKUOEGTSESUFQWXLJCJAUSRAYWUKUHVJVDNSVJDDDFXAJLCFVMJFXKVWUVVLUK
QWDTYMXJSFQKQYSJDA
2. UEHBCCFONNXNAOYRGTFNMDITNTJEZOCTHGNWTSUEQKJFMPCTNFWTCOEWLNEZTXIGTINBCVEISLGIFYFNUEYIHS
ATYOYNSFCCHTNIIDUO
3. GKEFEFAIIRSVYIDFUZEZIVNIEYOICINZMEOVHKTRORLZCRLPRCTYMINFOVIVORAXEJWXAFIUCJEKNEUTNWCMD
NFMUSZEZEPDIETOTZE

Key length 4. 78 letters each.

1. CEETCAQNSHODCTEJDNQTOEONEGOQSTCKLUPLYFTUONVLIVTADTWJCIVSEXIUECUSNDOSJFEQTESDT
2. UKECFNTRQAIYGZMMVKNYOZISHEFWKLQAMRANCKCIUWVNZRDGJABFFIJFYEWUWLHDWYUMSZFHIYIZA
3. GEBEBOIDNYCRUFNICTEUECTTMUTHXEOFCRTRUTMJEOSEODIEXNACECJGNVNNVIEQTMYNESCDQIODO
4. FHFKFIGXVXYFDFZLIIKJIGCZSNVWURJJZSCPWWYHUFDNVJXXFIXLVUMLKKFTVYMKAFITYZXC PKNTJUE

Key length 5. The first two groups have 63 letters; the last three have 62.

1. CKBKANDVOXUFJVTKOZTZGFHUCRFSNLNUYUONVRIFWBCUSFNFUWIKNYYZFFDNSZO
2. UEFTIXHIREMCITOCCEETWTQCZPARWUIEDIZDXTAADVJGKUUVMSWMYJZCKEIDE
3. GHTFOGSACFTMIIQYEGNHUVSLOJURTWTCJFLNOXDJNLIIJKIWN YUDTTSSSPTYOU
4. FECBINQYYCZNLNUIOSMNQKEJLMRPFKMUVVEJAGXXCFCLXENVCHQFOMECQIIJT
5. EEEFQRNXDGFZDKEJEITSOWXRKACCYCTHNWSVTDEIJFEMEYVTELEADUNXEHQTDA

Key length 6. 52 letters each.

1. CETBQDHCCFJCQUETEUXCCURYUUVSVDDXJCVJXVUVUQDYJSQQSD
2. UHCFNXAYGFMINJZCHNWUQJMCNWCUNWZXGIBVILYFUYHAYYSCHNIU
3. GEEAISYDUEINEOCNMOHTOLCLRTMNOIOAEWAICENUNCENMSEDEOT
4. FEKTGQXXDMLKOGSSFWLJASAWKHUJNDFALFMFKWVLKWTMFXFYJA
5. EBCONNORTNDTTEOTGTSEKFTFTOELETTNCESGINEISTONFCTIDO
6. KFFIRVIFZZVIYIIZEVKKRZRPCYIFVVRXJXFUJKETWMDFUZZPITZE

The IC below is already multiplied by 26, and the reported value is the average of the k groups. Checked from $k = 3$ to $k = 20$.

Key length	Average IC	Note
3	1.298	
4	1.279	
5	1.224	
6	1.797	shortest clear peak
7	0.983	
8	1.351	
9	1.323	
10	1.484	
11	1.050	
12	1.727	multiple of 6
13	1.138	
14	1.123	
15	1.497	
16	1.355	
17	1.031	
18	1.912	multiple of 6
19	1.063	
20	1.430	

Lengths 6, 12, and 18 peak. If the true length is 6, grouping every 12 or 18 letters still puts only one key letter in each group, so those multiples stay high. The shortest clear peak is 6.

Group ICs for 3, 4, and 6:

$k = 3$		$k = 4$		$k = 6$	
Group	IC	Group	IC	Group	IC
1	1.218	1	1.446	1	1.922
2	1.345	2	0.970	2	1.353
3	1.330	3	1.532	3	1.961
		4	1.169	4	1.392
				5	2.412
				6	1.745
avg	1.298	avg	1.279	avg	1.797

At length 6 every group stays near or above English. At 3 and 4 some groups stay near 1. The groups do not need identical ICs: each one is a Caesar shift of English, so the values move around 1.73. The key length is 6.

Playfair: Pokémon name

Explanation:

OTP (first line): Pokémon name

Explanation:

OTP (second line): Pokémon name

Explanation:

Generative AI Disclosure